



IDENTIFICATION

CODE : GI-3-S1-EC-APM
ECTS : 4

HOURS

Cours : 0h
TD : 30h
TP : 32h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 66h
Travail personnel : 16h
Total : 82h

ASSESSMENT METHOD

1 x 1h50 = written exam

TEACHING AIDS

Course material

TEACHING LANGUAGE

French

CONTACT

MME SANDOZ-GUERMOND
Francoise :
francoise.sandoz-guermond@insa-lyon.fr
M. DUMITRESCU Emil :
emil.dumitrescu@insa-lyon.fr

AIMS

This course belongs to teaching unit Informatique et mathématiques décisionnelles (GI-3-S1-UE-IMAD) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 2)
- A2 Operate a model of a real or virtual system (Level 2)
- A4 Design a system that meets a set of specifications (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
- C3 Evaluating, prototyping and simulating a system (Level 2)

- * Develop an experimental approach (Level 2)
- * Identify, formulate and solve a problem of complex engineering (Level 2)
- the following School skills : specific GI
- * evaluate, prototype and simulate a system (Level 2)
- * size the hardware and / or software of a system (Level 2)
- the following transversal School skills
- * Work, learn, evolve in an autonomous way (Level 2)
- * Interact with others, work in team (Level 3)

which decline the following professional skills :

- * Design and size systems required for a given activity of production, service, distribution ... (Level 2)
- * Identify and correct discrepancies in a continuous improvement process (material, human and information systems) by the implementation of analytical and simulation tools (Level 2)
- * Ensure quality reporting through the establishment of appropriate indicators, based on a systemic view of organizations (matrix, silo, project) for any scope and any type of activity (Level 2)"

CONTENT

- Adopt a structured programming approach to solving a complex problem
- Choose the solution methods and data structures appropriate for the problem
- Build and analyze algorithms in a pseudo-language
- Translate a UML class diagram into Python
- Transcribe an algorithm written in a pseudo-language into a Python program

BIBLIOGRAPHY

V. FELEA, Introduction à l'informatique, Apprendre à concevoir des algorithmes, Ed. Vuibert, 2013
L. DEBRAUWER, Algorithmique, Ed. ENI, 2008
T.CORMEN, C. LEISERSON, R.RIVEST, C. STEIN, Introduction à l'algorithmie, Ed Dunod, 2002
DELANNOY C., Programmer en Java, Ed Eyrolles, 2001
OUSSALAH C., Ingénierie objet, Interéditions, 1997

PRE-REQUISITES

